

Lecture 11

- Current
- Conductor

- 1-2 Midterm Exam examples will be uploaded tomorrow
- Wednesday: In-class exercise
- Friday: Review session + Some exam examples

Materials

Depending on whether the material contains **free charges** that can readily move in response to an applied electric field, we classify them into two broad categories: dielectrics and conductors.

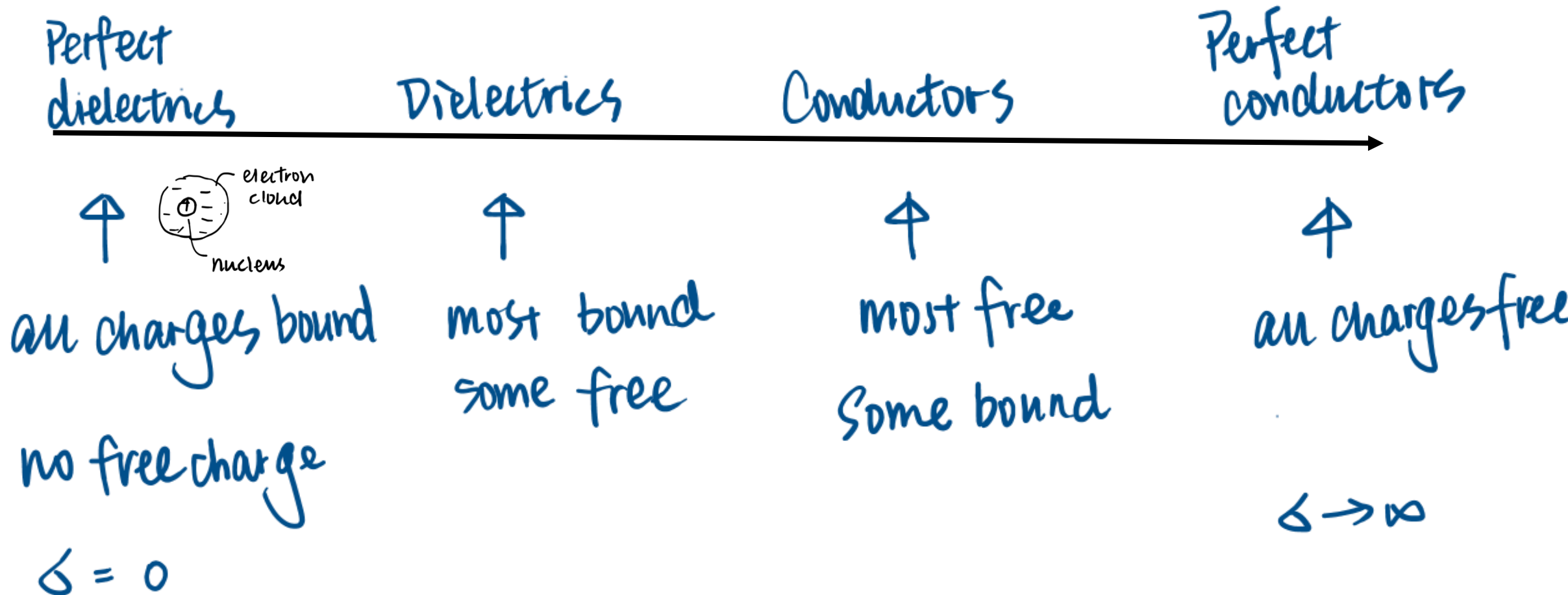
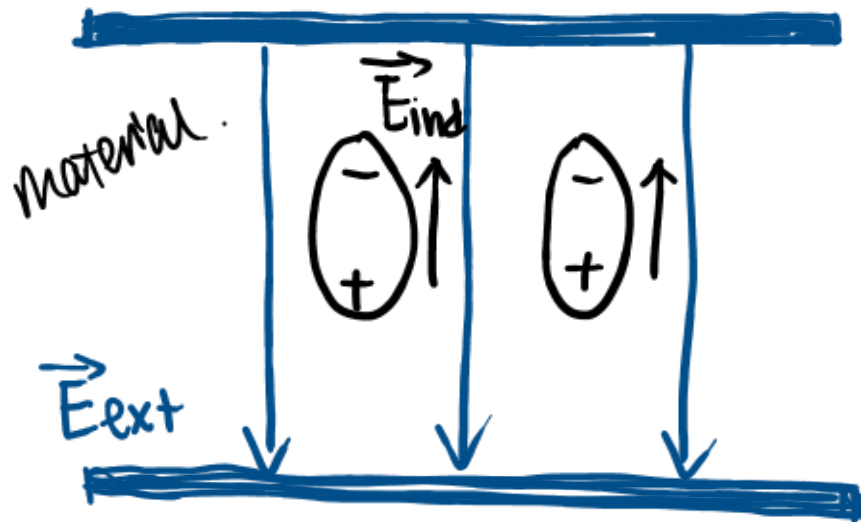
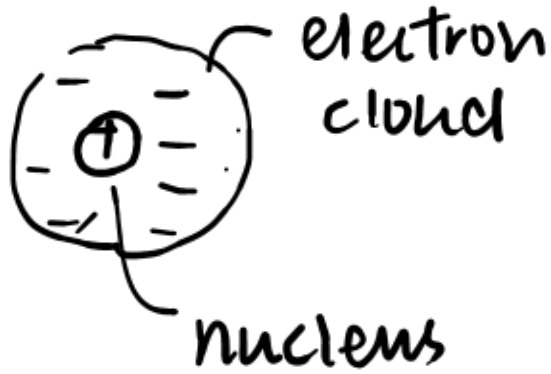


Table 4-1 Conductivity of some common materials at 20 °C.

Material	Conductivity, σ (S/m)
<i>Conductors</i>	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Iron	10^7
Mercury	10^6
Carbon	3×10^4
<i>Semiconductors</i>	
Pure germanium	2.2
Pure silicon	4.4×10^{-4}
<i>Insulators</i>	
Glass	10^{-12}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}

Review:

Perfect Dielectrics



$$\vec{E}_{\text{tot}} = \vec{E}_{\text{ext}} + \vec{E}_{\text{ind}}$$

$$\vec{E}_{\text{ext}} = \vec{E}_{\text{tot}} - \vec{E}_{\text{ind}}$$

free charge

bound charge

$$\epsilon_0 \vec{E}_{\text{ext}} = \epsilon_0 \vec{E}_{\text{tot}} - \epsilon_0 \vec{E}_{\text{ind}}$$

$$\vec{D} = \epsilon_0 \vec{E}_{\text{tot}} + \vec{P}$$

free charge

bound charge

Polarization Field

$\vec{P} = \chi_e \epsilon_0 \vec{E}_{\text{tot}}$

electron susceptibility

$\nabla \cdot \vec{D} = \rho_f$

affected by both free charge and bound charge

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E}_{\text{tot}}$$

$$\vec{D} = \epsilon_0 \epsilon_r \vec{E}_{\text{tot}}$$

ϵ_r : relative permittivity

$$\vec{D} = \epsilon \vec{E}_{\text{tot}}$$

$\left[\frac{C}{m^2} \right]$

$\left[\frac{F}{m} \right]$

$\left[\frac{V}{m} \right]$

Current

$\ominus - e$

Inside of metal, electrons are the only free charges.

So the total current is carried by electrons alone.

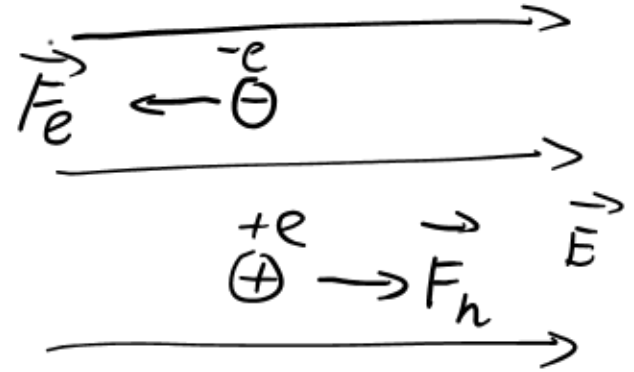
$\ominus - e$

$\oplus + e$

In semiconductors, there are both electrons and holes,

they function as independent channels, with total current

being the sum of electron and hole currents.



A hole is a missing electron in a semiconductor crystal that behaves like a positive charge carrier.

How fast do charge carriers move in a material?

Under an external electric field:

- If a charge carrier (electron or hole) were in vacuum, it would continue to accelerate, and its velocity would increase with time.
- In a conductor or semiconductor, charge carriers undergo:
 - Acceleration** due to the external electric field, and
 - Scattering** due to interactions with the crystal lattice, impurities, defects, etc.

At the macroscopic level, for a large number of carriers, the **average velocity of the charge carriers (drift velocity)** becomes effectively constant under a constant electric field, due to the balance between acceleration by the electric field and scattering.

⊖
electron drift velocity

$$\vec{v}_e = -\mu_e \vec{E}$$

↑
electron mobility

$$\left[\frac{m}{s} - \frac{m}{V} \right] = \left[\frac{m^2}{V \cdot s} \right]$$

⊕
hole drift velocity

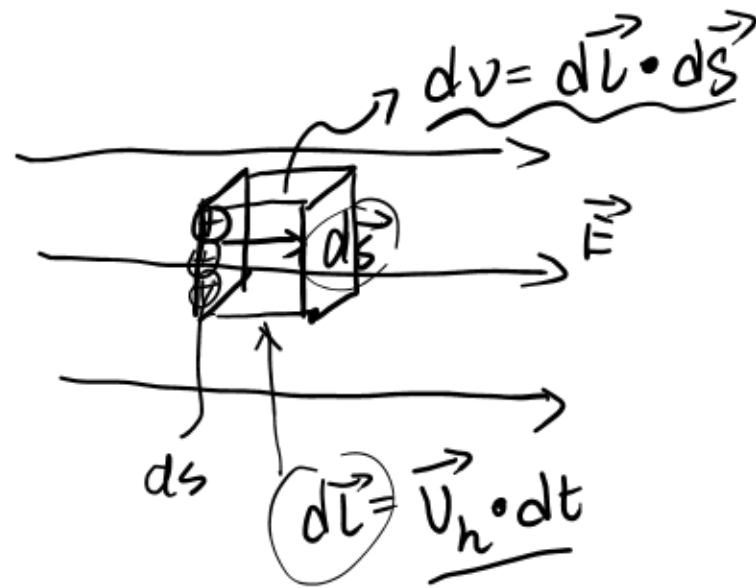
$$\vec{v}_h = \mu_h \vec{E}$$

↑
hole mobility.

Current is the rate at which charge flows through a surface.

Define inside of a material.

N_h : # [⊕]e holes/unit volume



$$dQ = N_h \cdot e \cdot dv$$

$$dQ = N_h e \cdot d\vec{l} \cdot d\vec{S}$$

$$= N_h e \cdot dt \cdot \vec{v}_h \cdot d\vec{S}$$

$$= N_h \cdot e \cdot dt \cdot \mu_h \cdot \vec{E} \cdot d\vec{S}$$

$$dI_h = \frac{dQ}{dt} = N_h \cdot e \cdot \mu_h \cdot \vec{E} \cdot d\vec{S}$$

$$dI = (N_h \cdot e \cdot \mu_h + N_e \cdot e \cdot \mu_e) \vec{E} \cdot d\vec{S}$$

$$dI = \frac{dQ}{dt}$$

dQ amount of charge flows thru a surface over dt .

N_e : # [⊖]electrons/unit volume.

$$dQ = N_e \cdot (-e) \cdot dv_e$$

$$= N_e (-e) \cdot \vec{v}_e \cdot dt \cdot d\vec{S}$$

$$= N_e (-e) \cdot (-\mu_e) \cdot \vec{E} \cdot dt \cdot d\vec{S}$$

$$dI_e = \frac{dQ}{dt} = N_e \cdot e \cdot \mu_e \cdot \vec{E} \cdot d\vec{S}$$

Current is the rate at which charge flows through a surface.

siemens

Define $\left[\frac{A}{V}\right] = [S] = \left[\frac{1}{\Omega}\right]$

$$dI = (N_h \cdot e \cdot \mu_h + N_e e \mu_e) \vec{E} \cdot d\vec{s}$$

depend on the material

||
conductivity σ
sigma

$$\left[\frac{A \cdot m}{V \cdot m^2}\right] = \left[\frac{A}{V \cdot m}\right] = \left[\frac{S}{m}\right] = \left[\frac{1}{\Omega \cdot m}\right]$$

* Ohm's law (differential form)

$$dI = \boxed{\sigma \cdot \vec{E} \cdot d\vec{s}}$$

current density $\left[\frac{A}{m^2}\right]$

$$\boxed{\vec{J} = \sigma \vec{E}}$$

* $\vec{J}$ is a vector.
BUT! I is a scalar with a sign $\begin{cases} + \\ - \end{cases}$

"flux"

$$\underline{dI} = \vec{J} \cdot d\vec{s}$$
$$I = \int_S dI = \int_S \vec{J} \cdot d\vec{s}$$

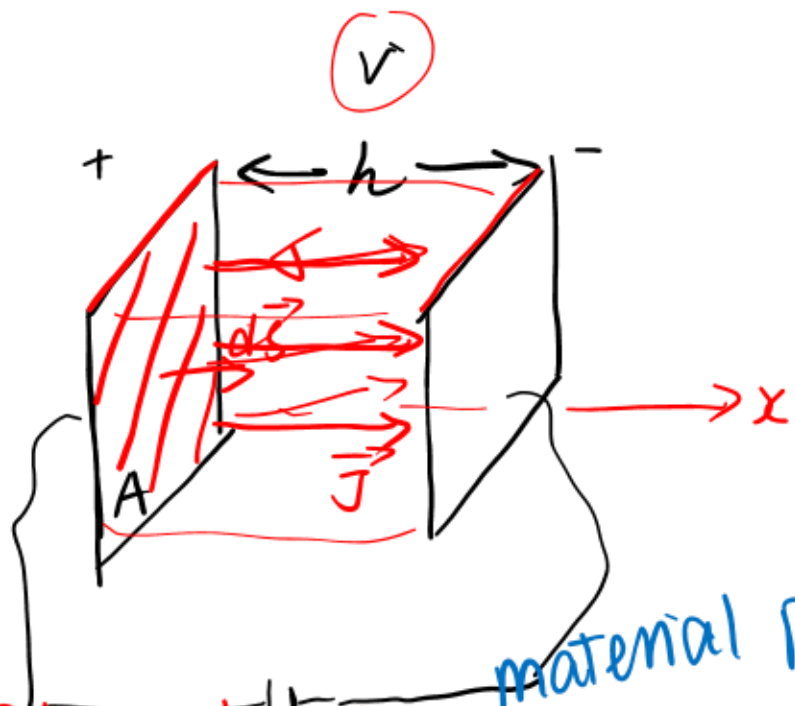


S: cross section

Ohm's law

Circuit: $V = IR$
 $\downarrow$
 resistance $[\Omega]$

$I = V \cdot G$
 $\downarrow$
 conductance $[S]$



$$V_{+-} = V_+ - V_- = - \int_-^+ \vec{E} \cdot d\vec{l} = \int_+^- \vec{E} \cdot d\vec{l} = E \cdot h$$

$$\vec{E} = \hat{x} \frac{V}{h}$$

$$\vec{J} = \sigma \vec{E} = \hat{x} \frac{\sigma V}{h}$$

$$I = \frac{\sigma V}{h} \cdot A$$

device property
 material + geometry of component

$$G = \frac{I}{V} = \frac{\sigma A}{h}$$

$$R = \frac{V}{I} = \frac{h}{\sigma A}$$

integral form

$$I = G \cdot V \rightarrow \text{conductance } [S]$$

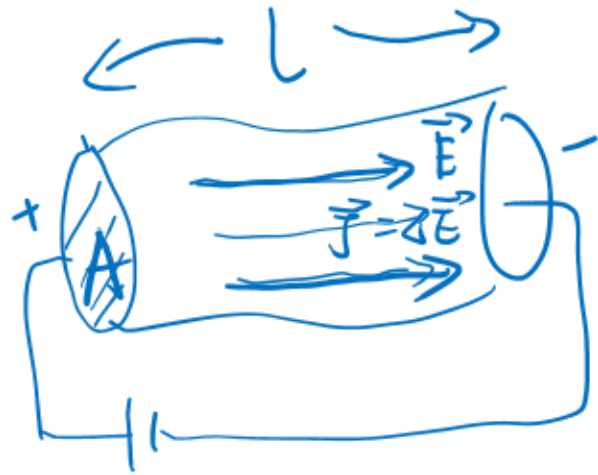
$$V = R \cdot I \rightarrow \text{resistance } [\Omega]$$

Ohm's law
 Differential form $\downarrow$
 $\vec{J} = \sigma \vec{E}$
 $\vec{E} = \rho \cdot \vec{J}$
 $\downarrow$
 conductivity $[\frac{S}{m}]$
 resistivity $[\Omega \cdot m]$

$$\sigma = \frac{1}{\rho}$$

Generalize expression of resistance R of certain device
conductance G

$$R = \frac{V}{I} = \frac{\int_{+}^{-} \vec{E} \cdot d\vec{l}}{\int_A \langle \vec{E} \rangle \cdot d\vec{s}} \quad [\Omega]$$



$$G = \frac{1}{R} \quad [S]$$

Perfect conductor (PEC)

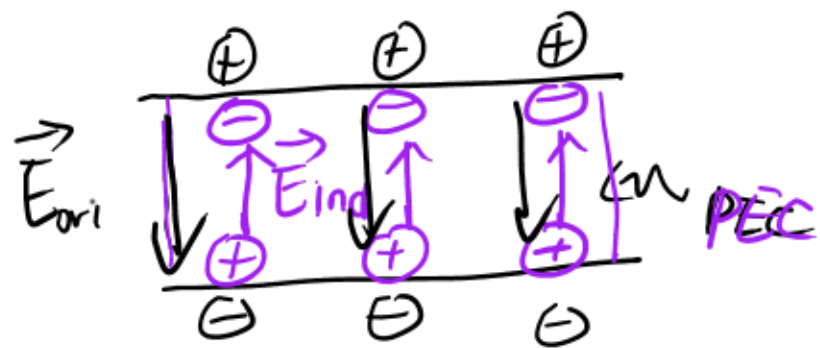
$$(\sigma \rightarrow \infty)$$

$$\vec{J} = \sigma \vec{E}$$

finite $\rightarrow \infty$

$$\Rightarrow \vec{E} = \frac{\vec{J}}{\sigma} = 0$$

Electric field inside of PEC: $\vec{E} = 0$

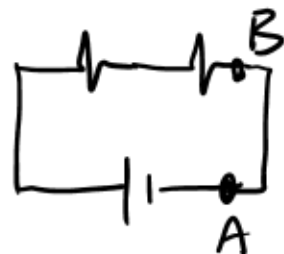


$$\vec{E}_{tot} = \vec{E}_{ori} + \vec{E}_{ind} = 0$$

Consequences:

① PEC is equipotential surface

② $\vec{E}_{external}$ is normal/ $\perp$ to PEC surface.

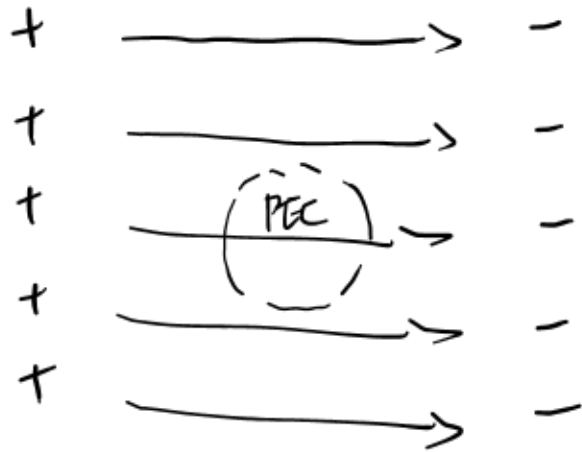


$$V_A = V_B$$

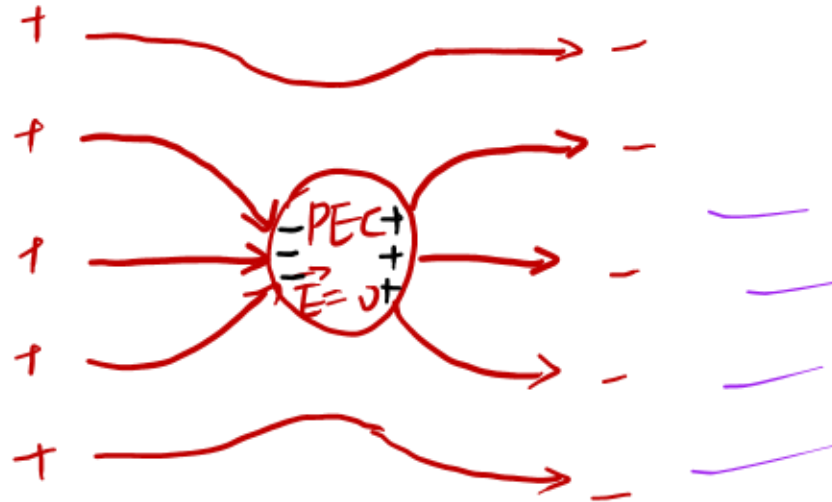
$$V_{AB} = V_A - V_B = -\int_B^A \vec{E} \cdot d\vec{L} = 0$$

Perfect conductor (PEC)

"Distorted"



$\Rightarrow$



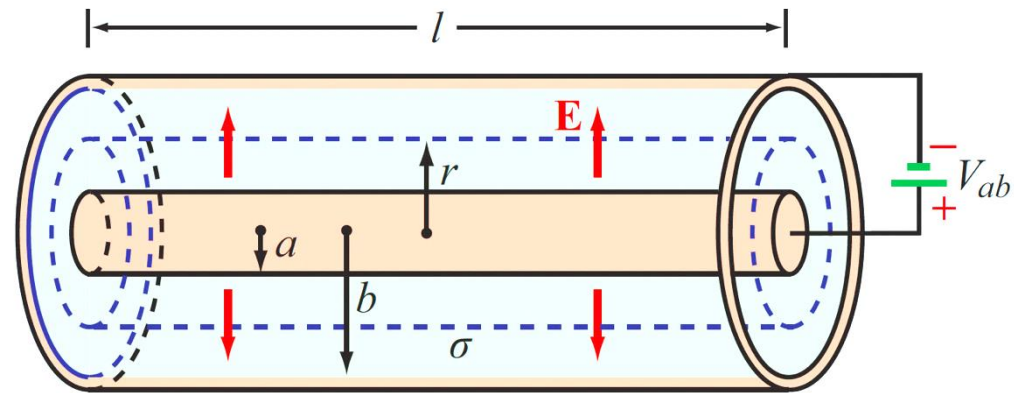
$\Rightarrow$ Within PEC net volume charge density ρ_v is 0 everywhere.

$$\Rightarrow \nabla \cdot \vec{D} = \rho_v$$

In the presence of an external electric field,
good dielectrics polarize,
good conductors screen/shield.

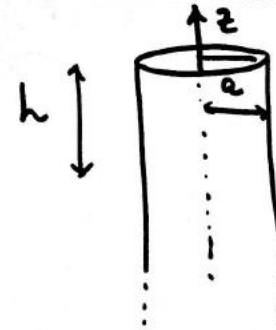
The free charges distribute themselves to the surface of PEC.
giving rise to surface charge density ρ_s .

The radii of the inner and outer conductors of a coaxial cable of length l are a and b , respectively (**Fig. 4-17**). The insulation material has conductivity σ . Obtain an expression for G' , which is the conductance per unit length of the insulation layer.



Problem 5:

a) A cylindrical shell of line charge density ρ_l has radius a . Use Gauss's law to calculate the electric field $\mathbf{E}$ everywhere in space. Assume the surrounding medium is air (permittivity ϵ_0).



b) Another cylindrical shell, with the line charge density of $-\rho_l$ and radius $b > a$ is placed coaxially around the shell from part a). Based on your result from a) and using the principle of superposition, calculate and graph the total electric field everywhere in space.



